

Implications for Computing and Quantum Error Correction

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Anchor: *Prime Valuation Depth: Multiplication as Branching, the Calculus of Indications, and the Structural No-Cloning Reading* (QNFO.RES.005, DOI 10.5281/zenodo.21918838).

Abstract

The anchor paper introduced the *branch-depth reading*: the p-adic valuation $v_p(n)$ is a measure of depth along a prime branch of the integer factorization tree (not a measure of size), and because the tensor product multiplies Hilbert-space dimensions, $v_2(\dim H) = n$ counts tensor-branch depth for n qubits. This follow-on subjects that reading to its own stated falsifiability condition in the two domains the anchor flagged but did not pursue — computing and quantum error correction. The central finding is a *correction, not a confirmation*: the naive mapping of a $[[n,k,d]]$ stabilizer code to branch depths, $n = v_2(\dim H)$ and $k = v_2(\dim H_L)$, is **trivially true by definition** (a code is defined on n qubits encoding 2^k logical states) and therefore carries no new content; and the code distance d — a minimum operator *weight*, not a *depth* — does not admit a valuation reading at all. The genuinely open question is whether there exists a **non-trivial** valuation invariant of stabilizer codes beyond the definitional n and k , and it is here — not in the naive mapping — that the existing QNFO "83% classification accuracy" claim (Kodaira–Néron classifier, DOI 10.5281/zenodo.21193487) lives and must be independently reproduced. The structural no-cloning reading is retained but repositioned honestly: it is a known categorical result (Abramsky 2009; Coecke 2009), which the branch-depth vocabulary re-expresses rather than discovers. Every non-empirical claim below carries an explicit falsifiability condition.

1. Introduction and Positioning

1.1 What the anchor paper established

Prime Valuation Depth developed three claims, each with a stated falsifiability condition:

1. **[TERRITORY — established, Ostrowski 1916]** Every positive integer is a finite product of prime powers; $v_p(n)$ is depth along the prime- p branch, not size.
2. **[MAP — interpretive]** The valuation-as-depth reading is a bridge between the calculus of indications (branching distinctions) and number theory (the prime divisor tree).

3. **[MAP — interpretive]** The tensor product multiplies dimensions, so prime factorization of $\dim H$ labels branch types and $v_p(\dim H)$ counts branch depth; the no-cloning theorem is then the impossibility of a linear diagonal map — cloning is nonlinear in the amplitudes, and quantum evolution preserves only linear structure.

1.2 The honest starting point of this paper

The anchor paper labels its own quantum reading as "interpretive rather than new physics" and attaches the falsifiability condition: *disconfirmed if the reading yields no explanatory or predictive content beyond the standard formalism — i.e., if it is pure relabeling*. This follow-on takes that condition seriously and applies it to the two domains the anchor flagged for later work. The result, reported here without flinching, is that **the most obvious extension — reading $[[n,k,d]]$ parameters as valuations — is largely pure relabeling**. This is not a failure of the program; it is the falsifiability machinery doing its job, and it sharpens the question to the one place where the branch-depth vocabulary could carry real weight.

1.3 External precedent (confirmation-bias correction)

The anchor's quantum reading does not occur in a vacuum. The categorical content — that quantum processes form a compact-closed (monoidal but non-Cartesian) category, and that no-cloning follows from this — is established in the categorical quantum mechanics literature [Abramsky & Coecke 2004; Abramsky 2009; Coecke 2009; Coecke & Duncan 2011]. Independently, the connection between p-adic geometry and quantum error-correcting codes is established in the holographic tensor-network literature [Heydemann, Marcolli, Saberi & Stoica 2018; Bhattacharyya, Hung, Lei & Li 2018; Gubser & Knaute 2017]. The present paper does not claim priority on either bridge; its narrow contribution is the specific *branch-depth vocabulary* (valuation as depth, in the calculus-of-indications sense) applied to code parameters, and its consequence for how QEC limits are framed.

2. The Branch-Depth Reading, Restated

Let $n \in \mathbb{Z}^+$ factor as $n = \prod p_i^{a_i}$. The anchor's reading: each prime p_i is a *branch type*, and the exponent $a_i = v_{p_i}(n)$ is the *depth* of nesting along that branch. A tensor product of Hilbert spaces multiplies dimensions: $\dim(H_1 \otimes H_2) = \dim(H_1) \cdot \dim(H_2)$. Hence for n qubits, $\dim H = 2^n$ and

$$v_2(\dim H) = n.$$

This is the single quantitative bridge the anchor supplies. The question this paper asks is: **what does it buy us?**

3. The $[[n,k,d]]$ Mapping — and Why It Is Mostly Relabeling

3.1 The definitional facts

A $[[n,k,d]]$ stabilizer code is a 2^k -dimensional subspace (the codespace) of an n -qubit Hilbert space $H \cong (\mathbb{C}^2)^{\otimes n}$, with $\dim H = 2^n$. Two identities follow immediately:

$$n = v_2(\dim H) = v_2(2^n) = n, \quad k = v_2(\dim H_L) = v_2(2^k) = k.$$

Claim [MAP — to be tested]: these identities constitute a "branch-depth reading" of the code parameters.

Finding [SELF-CORRECTION]: the identities are *definitional*. n is the number of qubits *by construction*, and k is the number of logical qubits *by construction*. Restating them as v_2 of a dimension adds nothing: the valuation is doing no work that the exponent of 2 in " 2^n " and " 2^k " was not already doing. This is the anchor's own falsifiability condition, triggered: **the n and k mappings are pure relabeling.**

3.2 The distance d does not have a valuation reading

The code distance d is the minimum weight (number of non-identity Pauli factors) of a non-trivial logical operator. It is a **Hamming weight** — a count of tensor factors an error touches — not a p -adic depth. There is no integer whose valuation yields d in the way 2^n yields n . The tempting phrase " d = branch-crossing error weight" is a metaphor, not a valuation identity, and must not be presented as mathematics. **[Falsifiability condition: disconfirmed if any specific valuation identity for d is claimed; d is a weight, and the burden of proof for any valuation-based bound on d is on the claimer.]**

3.3 What is left after the relabeling is stripped away

Stripping the definitional n and k mappings and the unavailable d mapping leaves a single, precise, and genuinely open question:

Does there exist a non-trivial valuation invariant of a stabilizer code — one not equal to $v_2(\dim H)$ or $v_2(\dim H_L)$ — that carries classification or predictive power across code families?

This is where the existing QNFO result attaches.

4. Computing as Path-Tracing: A Scoped, Falsifiable Claim

The anchor's computing leg was left as a flagged direction. The honest scope is narrow:

Claim [MAP — interpretive, scoped]: reversible classical computation and Clifford quantum computation are *path-tracing through a branching state space*, and the p -adic valuation is the natural depth

coordinate on that tree, in the specific sense that the Hensel-code arithmetic of exact rational computation [cf. the p-adic arithmetic literature] is computation *on* the branch tree.

Falsifiability condition: disconfirmed if no computational task can be shown to have a *valuation-based* complexity characterization that differs from (or tightens) its standard characterization. The claim is advanced only as a research program, not as an established result. **[CONTESTED] — no such characterization has yet been produced; this section is explicitly promissory.]**

5. Structural No-Cloning and the Necessity of QEC

5.1 The honest attribution

The claim "cloning would require a nonlinear diagonal map, and quantum evolution preserves only linear structure" is a known categorical result. In the categorical quantum mechanics framework, a compact-closed category is **non-Cartesian**: the monoidal product $\otimes$ is not a categorical product, so there is no natural diagonal $\Delta: A \rightarrow A \otimes A$ that is linear (a morphism in the category). No-cloning is the physical statement of the absence of this linear diagonal [Abramsky 2009; Coecke 2009]. The anchor paper's contribution is the *vocabulary* — expressing the same fact as "multiplicative branching cannot be linearly duplicated" — not the *result*.

Claim [MAP — re-expression]: the branch-depth vocabulary re-expresses no-cloning as: *a branch (tensor factor) cannot be duplicated by a linear process, so redundancy — the resource QEC consumes — must be carried in non-orthogonal (entangled) configurations, never in clones.*

Falsifiability condition: disconfirmed if this re-expression cannot be shown to yield at least one new, checkable consequence for QEC limits that the standard no-cloning statement does not already imply. **[Risk: HIGH — this is the relabeling risk again, and it is flagged as such.]**

5.2 The one place the framing could bite

A candidate consequence, stated as a falsifiable hypothesis: **if redundancy is non-cloneable, then the overhead (physical-to-logical qubit ratio) of a QEC code is bounded below by a function of the code's own valuation structure.** Whether this bound is tighter than, equivalent to, or weaker than the quantum Singleton bound $n - k \geq 2(d-1)$ is the test. **[UNTESTED — no bound has been derived; deriving one, and comparing it to the Singleton bound, is the concrete task that would elevate this framing from metaphor to mathematics.]**

6. The 83% Classification Claim: Status and Reproduction Requirement

QNFO.UF reports that a Kodaira–Néron-fiber classifier achieves **83% classification accuracy** on quantum error-correcting code families, with $v_p^{\max} = 28$ for optimal codes versus $v_p^{\max} = 4$ for random ensembles [Number-Theoretic Ultrametric Foundations, DOI 10.5281/zenodo.21193487]. The program registry summarizes this as "p-adic valuations classify QEC codes at 83% accuracy."

Status [UNVERIFIED-INTERNAL]: this is an internal report. It has not, to this author's knowledge, been independently reproduced on a fresh, held-out test set with a stated leakage-control protocol. Until it is, the 83% figure is a claim to be tested, not a result to be built upon.

Reproduction requirement (this paper's P4 critical path): a fresh train/test split with (i) no shared instances between training and evaluation, (ii) a stated classifier and hyperparameter protocol, (iii) a baseline (random-classifier accuracy and majority-class accuracy) reported alongside, and (iv) the v_p -spectral invariant computed independently. Success is reproduction *within stated confidence*; failure is reported honestly either way.

Falsifiability condition: the "83% accuracy" claim is disconfirmed if independent reproduction fails to exceed a stated baseline by a pre-registered margin. **[The reproduction is out of scope for this vo.1 manuscript and is deferred to P4; its absence is disclosed, not hidden.]**

7. Complete Falsifiability Register

#	Claim	Type	Falsifiability condition	Current status
C1	$v_p(n)$ is depth along a prime branch	established (Ostrowski)	—	established
C2	$n = v_2(\dim H)$, $k = v_2(\dim H_L)$ is a "branch-depth reading" of $[[n, k, d]]$	MAP	disconfirmed (pure relabeling)	SELF-CORRECTED
C3	d admits a valuation reading	MAP	no valuation identity exists	REJECTED
C4	a non-trivial valuation invariant of codes exists	open question	no such invariant found $\Rightarrow$ program fails	OPEN
C5	computing = path-tracing, with valuation-based complexity content	MAP	no valuation-based complexity characterization produced	PROMISSORY
C6	no-cloning re-expressed as non-cloneable redundancy	re-expression	no new checkable consequence $\Rightarrow$ relabeling	RISK-HIGH
C7	overhead bounded by valuation structure	hypothesis	compare to Singleton bound	UNTESTED
C8	Kodaira–Néron classifier is 83% accurate	empirical (internal)	independent reproduction vs baseline	UNVERIFIED-INTERNAL

8. Relation to Prior Work (External)

- **Categorical QM / no-cloning:** Abramsky & Coecke (2004) *A categorical semantics of quantum protocols*; Abramsky (2009) *No-Cloning in Categorical Quantum Mechanics*; Coecke (2009) *Quantum Pictorialism*; Coecke & Duncan (2011) *Interacting Quantum Observables*. The no-cloning content of §5 is theirs; this paper adds only vocabulary.
- **p-adic holographic QEC:** Heydeman, Marcolli, Saberi & Stoica (2018) *Tensor networks, p-adic fields, and algebraic curves* (ATMP 22:93); Bhattacharyya, Hung, Lei & Li (2018) *Tensor network and (p-adic) AdS/CFT*; Gubser & Knaute (2017) *p-Adic AdS/CFT*. The p-adic/QEC connection is theirs; this paper's branch-depth framing is a distinct, narrower lens.
- **Stabilizer methods:** Bravyi, Browne, Calpin, Campbell, Gosset & Howard (2019) *Simulation of quantum circuits by low-rank stabilizer decompositions* — the working background for §3–§4.

Relation to QNFO internal corpus: the p-adic QEC space is densely worked internally (see artifacts/due-diligence.md §2 for the DOI list). This paper's narrow differentiation is the branch-depth vocabulary and the self-correction of §3, which the existing corpus (which treats valuation as a *classifier weight*, not a *depth*) does not perform.

9. Conclusion

The branch-depth reading, applied honestly to computing and quantum error correction, yields a **negative and a positive result**. The negative result: the obvious extension — reading $[[n,k,d]]$ parameters as valuations — is definitional for n and k and unavailable for d ; it is, in the anchor's own terms, pure relabeling, and this paper says so rather than papering over it. The positive result: stripping the relabeling leaves one precise open question — *does a non-trivial valuation invariant of stabilizer codes exist?* — and one concrete falsifiable task — *derive a valuation-based QEC-overhead bound and compare it to the Singleton bound*. The 83% classification claim is a testable hypothesis at the center of that question, and it must be independently reproduced before it is treated as established. This is the honest state of the program: a sharp question, a clear test, and a prior claim awaiting verification.

References

1. Abramsky, S., & Coecke, B. (2004). A categorical semantics of quantum protocols. *LICS 2004*.
2. Abramsky, S. (2009). No-Cloning in Categorical Quantum Mechanics. In *Semantic Techniques in Quantum Computation*.
3. Coecke, B. (2009). Quantum Pictorialism. *Contemporary Physics*.
4. Coecke, B., & Duncan, R. (2011). Interacting Quantum Observables: Categorical Algebra and Diagrammatics. *New Journal of Physics*.
5. Heydeman, M., Marcolli, M., Saberi, I., & Stoica, B. (2018). Tensor networks, p-adic fields, and algebraic curves. *Adv. Theor. Math. Phys.* 22:93. arXiv:1605.07639.
6. Bhattacharyya, A., Hung, L.-Y., Lei, Y., & Li, W. (2018). Tensor network and (p-adic) AdS/CFT. *JHEP*. arXiv:1703.05445.
7. Gubser, S. S., & Knaute, J. (2017). p-Adic AdS/CFT. *Commun. Math. Phys.*
8. Bravyi, S., Browne, D., Calpin, P., Campbell, E., Gosset, D., & Howard, M. (2019). Simulation of quantum circuits by low-rank stabilizer decompositions. *Quantum* 3:181.
9. Ostrowski, A. (1916). Über einige Lösungen der Funktionalgleichung $\varphi(x)\varphi(y)=\varphi(xy)$. *Acta Mathematica*.
10. Quni-Gudzinis, R. B. (2026). Prime Valuation Depth. Zenodo. DOI 10.5281/zenodo.21918838.
11. QNFO Research Collective (2026). Number-Theoretic Ultrametric Foundations. Zenodo. DOI 10.5281/zenodo.21193487.